

Karl Svozil

Ao.Univ.Prof. i.R. Dr., TU Wien

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Professional Summary

Distinguished Theoretical Physicist and Emeritus Professor (Ao.Univ.Prof. i.R.) at the Vienna University of Technology (TU Wien). Extensively published researcher with over 190 peer-reviewed works specializing in quantum mechanics, theoretical physics, quantum contextuality, randomness, and computer science logic. Recognized international academic leader, former President of the International Quantum Structure Association, and frequent collaborator with global academic and scientific institutions.

Professional Experience

TU Wien (Vienna University of Technology)	<i>Vienna, Austria</i>
<i>Ao.Univ.Prof. (iR-Emeritus) / Professor of Theoretical Physics</i>	<i>1997 – Present</i>

- Conduct high-level, interdisciplinary research at the Institute for Theoretical Physics focusing on quantum contextuality, value indefiniteness, and advanced quantum information.
- Teach and mentor graduate students; co-organizer of the PhD program in Physics and Computer Science (since 2011).
- Transitioned to Emeritus status (retired Univ. Dozent) as of October 2022, while continuing active publication and peer review.

<i>Permanent Researcher</i>	<i>1990 – 1997</i>
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University of Auckland	<i>Auckland, New Zealand</i>
<i>Honorary Researcher (Computer Science)</i>	<i>Dec 2011 – Feb 2018</i>

- Held an honorary appointment at the Centre for Discrete Mathematics and Theoretical Computer Science.
- Co-authored pivotal papers on quantum random number generation and value indefinite observables with international researchers.

Lomonosov Moscow State University	<i>Moscow, Russia</i>
<i>Visiting Scholar (Physics)</i>	<i>Sep 1986 – Oct 1986</i>

Austrian Ministry for Science & Research	<i>Vienna, Austria</i>
<i>Staff Scientist</i>	<i>1984 – 1990</i>

Education

TU Wien (Vienna University of Technology)	<i>Vienna, Austria</i>
Universitätsdozent (Habilitation)	<i>Mar 1988</i>

University of California at Berkeley & Lawrence Berkeley Laboratory	<i>Berkeley, CA, USA</i>
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Visiting Scholar, Rotary Foundation Scholar in Theoretical Physics	<i>Oct 1982 – Jul 1983</i>
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Academic & Organizational Leadership

- **International Quantum Structure Association:** President (2012–2014) and Vice President (2014–2016).
- **Fonds Wetenschappelijk Onderzoek - Vlaanderen:** Panel Member for the Belgian research foundation (2010–Present).
- **Editorial Duties:** Associate Editor of multiple scientific journals in Physics and Computer Science (1994–Present). Active peer reviewer for *Entropy*, *Foundations of Physics*, *npj Quantum Information*, and *International Journal of Theoretical Physics*.
- **Global Academic Collaboration:** Conducted prolonged research visits at academic institutions across the USA, Canada, New Zealand, Germany, Russia, Italy, Denmark, Malaysia, and India.

Selected Recent & Notable Publications

- **Svozil, K.** (2026). “From Unitarity to Irreversibility: The Role of Infinite Tensor Products and Nested Wigner’s Friends”. *Foundations of Physics*, 56(4). [DOI: 10.1007/s10701-025-00903-9].
- **Navara, M., & Svozil, K.** (2025). “Exploring Quantum contextuality with the quantum Möbius-Escher-Penrose hypergraph”. *Physical Review A*, 111(4), 042209.
- **Calude, C. S., & Svozil, K.** (2024). “Binary quantum random number generator based on value indefinite observables”. *Scientific Reports (Nature)*, 14, 12845.
- **Svozil, K.** (2024). “Converting nonlocality into contextuality”. *Physical Review A*, 110, 012215.
- **Shekarriz, M. H., & Svozil, K.** (2022). “Noncontextual coloring of orthogonality hypergraphs”. *Journal of Mathematical Physics*, 63(3), 032104.
- **Svozil, K.** (2018). *Physical (A)Causality. Determinism, Randomness and Uncaused Events*. Springer, Cham. (Book).
- **Abbott, A. A., Calude, C. S., & Svozil, K.** (2014). “Value-indefinite observables are almost everywhere”. *Physical Review A*, 89(3), 032109.
- **Svozil, K., & Tkadlec, J.** (1996). “Greechie diagrams, nonexistence of measures in quantum logics and Kochen-Specker type constructions”. *Journal of Mathematical Physics*, 37, 5380-5401.

For a comprehensive list of 190+ works, please refer to ORCID: 0000-0001-6554-2802.